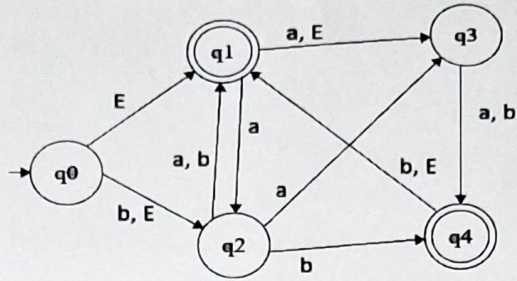


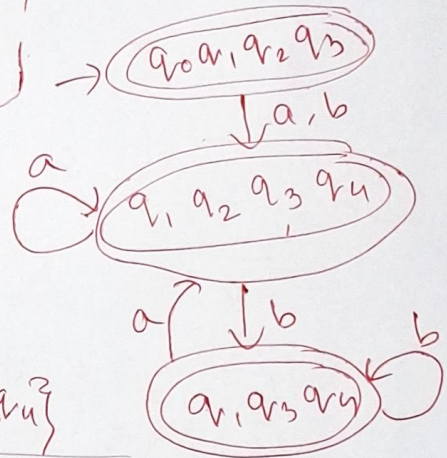
U 1000-... U 1010-... U 1001-...

2. (10 marks) Convert the following NFA into equivalent DFA.



	a	b
$\{q_0, a_1, a_2, a_3\}$	$\{q_3, q_2, a_1, a_4\}$	$\{q_2, q_4, a_1, a_3\}$
$\{a_1, a_3\}$	$\{q_2, a_3, a_4, a_1\}$	$\{a_1, a_3, a_4\}$
$\{a_2\}$	$\{a_1, a_3\}$	$\{a_4, a_1, a_3\}$
$\{a_3, a_4\}$	$\{a_4, a_1, a_3\}$	$\{a_4, a_1, a_3\}$
$\{a_4, a_1, a_3\}$	$\{a_1, a_4, a_3\}$	$\{a_1, a_4, a_3\}$

correction



	a	b
$\{q_0, a_1, a_2, a_3\}$	$\{a_1, a_2, a_3, a_4\}$	$\{a_1, a_2, a_3, a_4\}$
$\{a_1, a_2, a_3, a_4\}$	$\{a_1, a_2, a_3, a_4\}$	$\{a_1, a_3, a_4\}$
$\{a_1, a_3, a_4\}$	$\{a_1, a_2, a_3, a_4\}$	$\{a_1, a_3, a_4\}$

Solution

CSE331 - Automata and Computability

Quiz 1

Section __ (Set B)

Date:

Full name:

Roll:

Total Marks: 20

1. (10 marks) Write Regular expression, $\Sigma = \{0, 1\}$

$L_1 = \{w \mid \text{length of } w \text{ is at most } 3\}$

$L_2 = \{w \mid \text{length of } w \text{ is at least } 3\}$

$L_3 = \{w \mid \text{contains } 01 \text{ as subsequence}\}$

a. Write a regular expression for language L_1 , L_2 and L_3

b. Write RE for $\overline{L_1}$, $\overline{L_2}$, and $\overline{L_3}$

c. Write RE for $\overline{L_1 \cup L_3}$ and $\overline{L_2 \cap L_3}$.

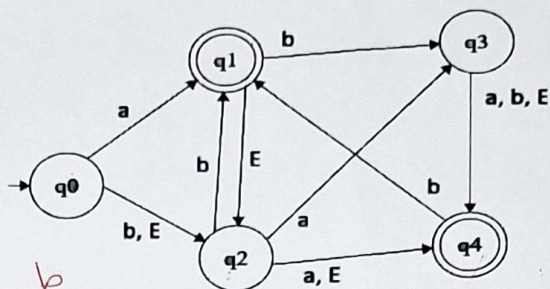
a $L_1 = \epsilon \cup \Sigma \cup \Sigma\Sigma \cup \Sigma\Sigma\Sigma$
 $L_2 = \Sigma\Sigma\Sigma\Sigma^*$
 $L_3 = \Sigma^*0\Sigma^*1\Sigma^*$

b $\overline{L_1} = \Sigma\Sigma\Sigma\Sigma^* \text{ (at least 4)}$
 $\overline{L_2} = \epsilon \cup \Sigma \cup \Sigma\Sigma \text{ (at most 2)}$
 $\overline{L_3} = \Sigma^*0\Sigma^*0\Sigma^* \cup \Sigma^*1\Sigma^*1\Sigma^* \cup \Sigma^*0\Sigma^*1\Sigma^*0\Sigma^* \cup \Sigma^*1\Sigma^*0\Sigma^*1\Sigma^*0\Sigma^*$

c $\overline{L_1 \cup L_3} = \overline{L_1} \cap \overline{L_3} = 00000^* \cup 11111^*0^* \cup 10000^* \cup 11000^* \cup 11100^*$

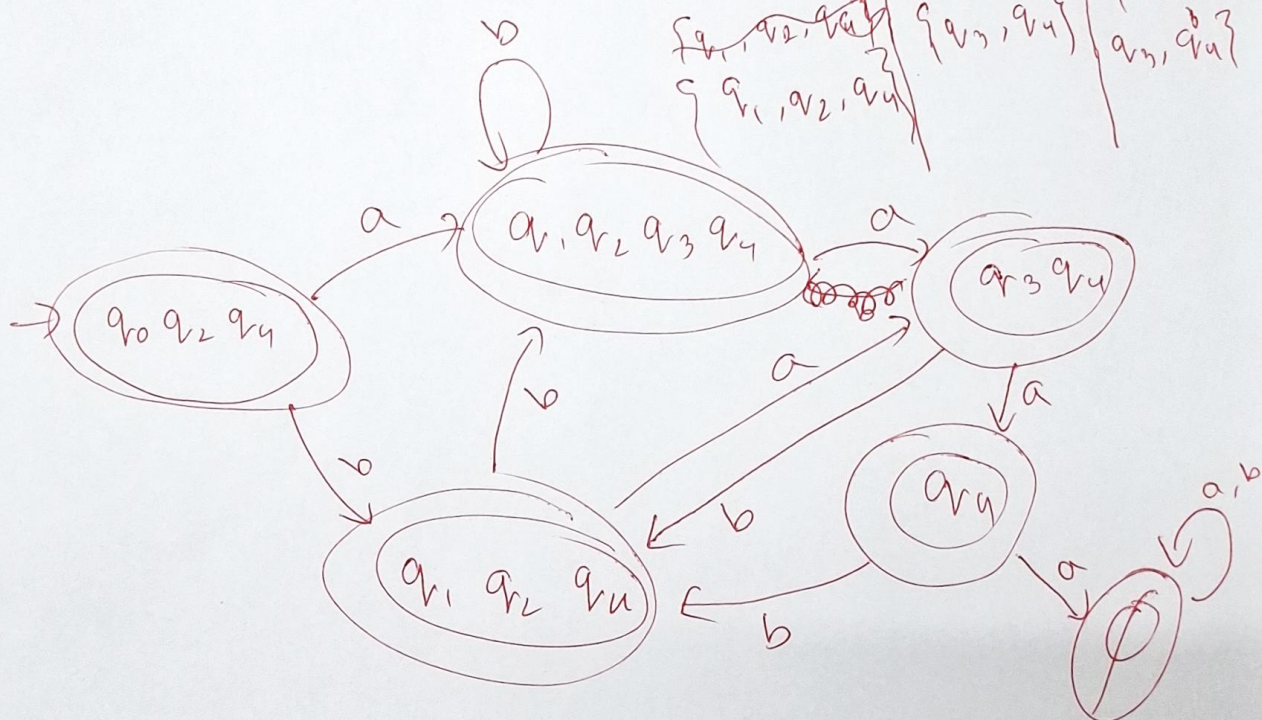
$\overline{L_2 \cap L_3} = \overline{L_2} \cup \overline{L_3} = \epsilon \cup \Sigma \cup \Sigma\Sigma \cup \Sigma^*0\Sigma^*0\Sigma^* \cup \Sigma^*1\Sigma^*1\Sigma^*0\Sigma^* \cup \Sigma^*0\Sigma^*1\Sigma^*0\Sigma^* \cup \Sigma^*1\Sigma^*0\Sigma^*1\Sigma^*0\Sigma^*$

2. (10 marks) Convert the following NFA into equivalent DFA.



	a	b
$\{q_0, q_2, q_4\}$	$\{q_1, q_2, q_3, q_4\}$	$\{q_2, q_4, q_1\}$
$\{q_1, q_2\}$	$\{q_3, q_4\}$	$\{q_3, q_4, q_1, q_2\}$
$\{q_2, q_4\}$	$\{q_3, q_4\}$	$\{q_1, q_2, q_4\}$
$\{q_3, q_4\}$	$\{q_4\}$	$\{q_4, q_1, q_2\}$
$\{q_4\}$	$\emptyset$	$\{q_1, q_2, q_4\}$

	a	b
$\{q_1, q_2, q_3, q_4\}$	$\{q_1, q_2, q_3, q_4\}$	$\{q_1, q_2, q_4\}$
$\{q_3, q_4\}$	$\{q_3, q_4\}$	$\{q_1, q_2, q_3, q_4\}$
$\{q_4\}$	$\emptyset$	$\{q_1, q_2, q_4\}$
$\emptyset$	$\emptyset$	$\{q_1, q_2, q_4\}$
$\{q_1, q_2, q_4\}$	$\{q_3, q_4\}$	$\{q_1, q_2, q_4\}$



CSE331 - Automata and Computability

Quiz 2

Section __ (Set C)

Date:

Full name:

Roll:

Total Marks: 20

1. (10 marks) Write Regular expression, $\Sigma = \{0, 1\}$

$L_1 = \{w \mid \text{length of } w \text{ is at least } 3\}$

$L_2 = \{w \mid \text{starts with } 00\}$

$L_3 = \{w \mid \text{contains } 11 \text{ as a substring}\}$

a. Write a regular expression for language L_1 , L_2 and L_3

b. Write RE for $\overline{L_1}$, $\overline{L_2}$, and $\overline{L_3}$

c. Write RE for $\overline{L_1 \cap L_3}$ and $\overline{L_2 \cup L_3}$.

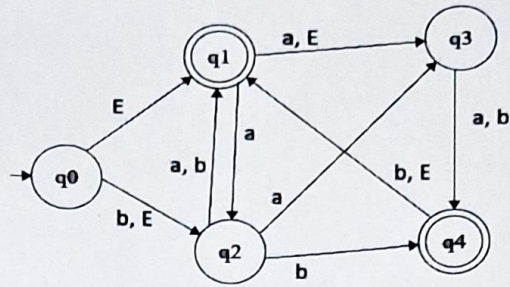
a) $L_1 = \Sigma \Sigma \Sigma \Sigma^*$
 $L_2 = 00 \Sigma^*$
 $L_3 = \Sigma^* 11 \Sigma^*$

b) $\overline{L_1} = \epsilon \cup \Sigma \cup \Sigma \Sigma \text{ (at most 2)}$
 $\overline{L_2} = 01 \Sigma^* \cup 10 \Sigma^* \cup 11 \Sigma^* \cup 00 \cup 1 \cup \epsilon$
 $\overline{L_3} = \epsilon \cup 00 \cup 1 \cup 0^* (10^+)^* (\epsilon \cup 1)$

c) $\overline{L_1 \cap L_3} = \overline{L_1} \cup \overline{L_3} = \epsilon \cup \Sigma \cup \Sigma \Sigma \cup 00 \cup 1 \cup 0^* (10^+)^* (\epsilon \cup 1)$

$\overline{L_2 \cup L_3} = \overline{L_2} \cap \overline{L_3} = \epsilon \cup 00 \cup 1 \cup 010^+ (10^+)^* (\epsilon \cup 1)$
 $\cup 100^+ (10^+)^* (\epsilon \cup 1)$

2. (10 marks) Convert the following NFA into equivalent DFA.



Same as set A

CSE331 - Automata and Computability

Quiz 2

Section __ (Set D)

Date:

Full name:

Roll:

Total Marks: 20

1. (10 marks) Write Regular expression, $\Sigma = \{0, 1\}$

$L_1 = \{w \mid \text{length of } w \text{ is at most } 3\}$

$L_2 = \{w \mid \text{starts with } 11\}$

$L_3 = \{w \mid \text{contains } 01 \text{ as subsequence}\}$

a. Write a regular expression for language L_1 , L_2 and L_3

b. Write RE for $\overline{L_1}$, $\overline{L_2}$, and $\overline{L_3}$

c. Write RE for $\overline{L_1 \cup L_3}$ and $\overline{L_2 \cap L_3}$.

a)

$$L_1 = \epsilon \cup \Sigma \cup \Sigma\Sigma \cup \Sigma\Sigma\Sigma$$

$$L_2 = 11\Sigma^*$$

$$L_3 = \Sigma^* 0 \Sigma^* 1 \Sigma^*$$

b)

$$\overline{L_1} = \Sigma\Sigma\Sigma\Sigma^* (\text{at least } 4)$$

$$\overline{L_2} = 00\Sigma^* \cup 10\Sigma^* \cup 01\Sigma^* \cup 001 \cup \epsilon$$

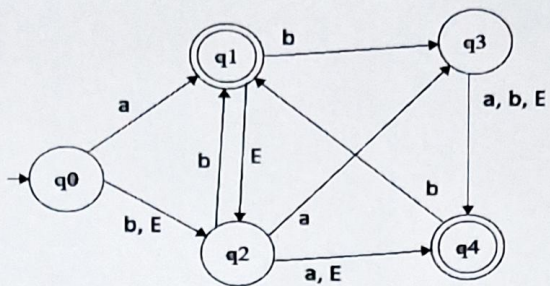
$$\overline{L_3} = \epsilon \cup 0^* \cup 1^* \cup 0^* 1^* 0^*$$

c)

$$\overline{L_1 \cup L_3} = \overline{L_1} \overline{L_3} = 00000^* \cup 11111^* 0^* \cup 10000^* \cup 11000^* \cup 11100^*$$

$$\overline{L_2 \cap L_3} = \overline{L_2} \overline{L_3} = 00\Sigma^* \cup 10\Sigma^* \cup 01\Sigma^* \cup 001 \cup \epsilon \cup 1^* 0^*$$

2. (10 marks) Convert the following NFA into equivalent DFA.



Same as set B